

Assignment 3

Question 1)

a)

option 1: NOR6 + INV

$$g_{NOR_n} = \frac{2n+1}{3} \quad p_{NOR_n} = n$$

For NOR6 $g_1 = \frac{13}{3}$ $p_1 = 6$

Inverter $g_2 = 1$ $p_2 = 1$

$$G = \frac{13}{3} \quad H = \frac{90}{30} = 3 \quad F = GH = 13$$

$$P = \sqrt{F} = 3.606$$

$$D = 2P + P = 2(3.606) + (6+1) = 14.21$$

option 2: NOR3, NAND2

$$F = (A+B+C) \cdot (D+E+G) = A+B+C \cdot D+E+G$$

For NOR3

$$g_{NOR3} = \frac{7}{3}, \quad p_{NOR3} = 3$$

For NAND2: $g_{NAND2} = \frac{4}{3}, \quad p_{NAND2} = 2$

$$G = \frac{7}{3} \cdot \frac{4}{3} = \frac{28}{9}$$

$$H = \frac{90}{30} = 3 \Rightarrow F = GH = \frac{28}{3} = 9.333$$

$$P = \sqrt{\frac{28}{3}} = 3.056$$

$$D = 2P + P = 2(3.056) + (3+2) = 11.11 \Rightarrow \text{faster}$$

so ~~for~~

~~for~~ for a unit NOR3 $\Rightarrow C_{in} = 6 + 1 = 7$

since each input is allowed 30 $\Rightarrow$ each NOR3 has a gate cap of 30

$$\Rightarrow \chi_{NOR3} = \frac{30}{7} = 4.286$$

each NOR3 has:

each NMOS: $1x = 4.291$

PMOS: $6x = 25.714$

$$\text{NAND2: } h_1 = \frac{f}{g_1} = \frac{3.096}{\frac{2}{3}} = 1.31$$

$$C_{in, NAND2} = h_1 C_{in} = 1.31 (30) = 39.3$$

Unit NAND2 input cap is $2 + 2 = 4$ so

$$\chi_{NAND2} = \frac{39.3}{4} = 9.82$$

each NAND2:

each NMOS: $2x = 19.641$

each PMOS $2x = 19.641$

b) ~~best topology~~

~~Pseudo-dynamic NOR6~~ ~~HT-stew inverter~~

Stage 1) best choice: Pseudo-nMOS NOR6 restoring inverter
using 1 always on pMOS load and six parallel nMOS
each input NMOS: 30k
load PMOS: 15k

Stage 2: restoring inverter

using $C_{in2} \approx 64$

$$\chi_{inv} = \frac{64}{3} = 21.3$$

inverter nMOS: 21.3 n

inverter pMOS: 42.7

$$D = 7.5$$

c) best topology: booted dynamic NOR6 + HI-skew inverter

each input nMOS 30 n

booter nMOS: 30 n

precharge pMOS: 15 n

Unit HI-skew inverter has input cap of $2+0.5 = 2.5$

dynamic NOR6 falling logical effort

$$g_1 = \frac{2}{3}$$

HI-skew inverter rising logical effort.

$$g_2 = \frac{5}{6}$$

$$G = \frac{2}{3} \cdot \frac{5}{6} = \frac{5}{9}$$

$$H = \frac{20}{3} = 3$$

$$F = \frac{5}{3} = 1.667$$

$$P = \sqrt{1.667} = 1.291$$

dynamic NOR6 output Node $P_1 = \frac{2(6)+1}{3} = \frac{13}{3}$

HI-skew Inverter $P_2 = \frac{2+0.5}{3} = \frac{2.5}{3}$

$$P = \frac{13}{3} + \frac{2.5}{3} = 5.167$$

$$D = 2P \cdot P = 7.75$$

$$h_1 = \frac{P}{g_1} = \frac{1.291}{\frac{2}{3}} = 1.936$$

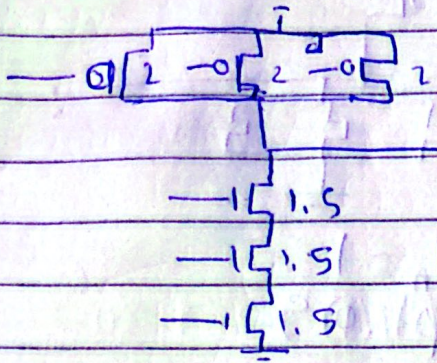
$$C_{in, HI} = 1.936(30) = 58.1$$

$$\chi_{HI} = \frac{58.1}{2.5} = 23.24$$

Question 2)

NAND:

HI-skew



$$W_p = 2$$

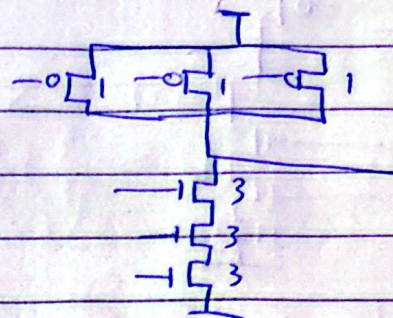
$$3 \cdot \frac{R}{W_n} = 2R \Rightarrow W_n = 1.5$$

$$C_{in} = W_p + W_n = 2 + 1.5 = 3.5$$

$$g = \frac{3.5}{3} = \boxed{\frac{7}{6}}$$

SI/O

Low skew



$$3 \cdot \frac{R}{W_n} = R \Rightarrow W_n = 3$$

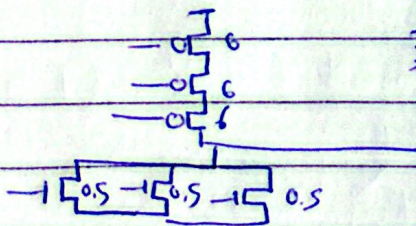
$$2 \frac{R}{W_p} = 2R \Rightarrow W_p = 1$$

$$C_{in} = 1 + 3 = 4$$

$$g = \boxed{\frac{4}{3}}$$

NOR

HI-skew



$$3 \cdot \frac{2R}{W_p} = 12$$

$$W_p = 6$$

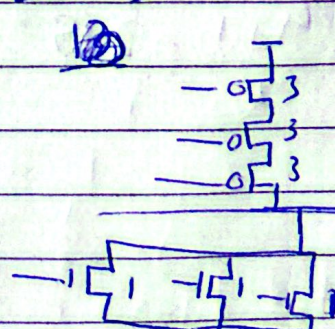
$$\frac{R}{W_n} = 2R$$

$$W_n = 0.5$$

$$C_{in} = 6 + 0.5 = 6.5$$

$$g = \frac{6.5}{3} = \boxed{\frac{13}{6}}$$

Low-skew



$$\frac{R}{W_n} = R \Rightarrow W_n = 1$$

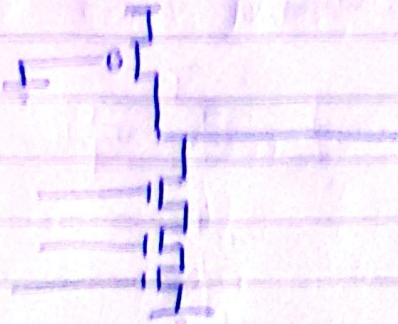
$$3 \cdot \frac{2R}{W_p} = 2R \Rightarrow W_p = 3$$

$$C_{in} = 3 + 1 = 4$$

$$g = \boxed{\frac{4}{3}}$$

Question 3)

NAND



$$W_n = 3$$

$$R_n = \frac{R}{3} + \frac{R}{3} + \frac{R}{3} = R$$

$$W_p = 0.5$$

$$C_{in} = 3$$

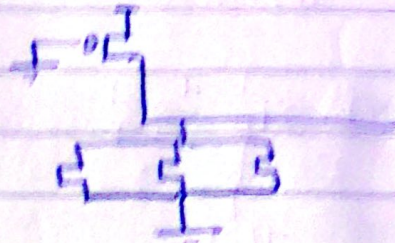
$$g_j = \frac{C_{in}}{3} = \frac{3}{3} = 1$$

$$C_{inv, rise} = 0.5 + 0.75 = 0.75$$

$$g_0 = \frac{3}{0.75} = 4$$

$$g_{avg} = \frac{4 + 1}{2} = 2.5$$

NOR



$$W_n = 1$$

$$W_p = 0.5$$

$$C_{in} = 1$$

$$g_j = \frac{1}{3}$$

$$g_0 = \frac{1}{0.75} = \frac{4}{3}$$

$$g_{avg} = \frac{9}{6}$$

Question 4)

$$AND_8 = \overline{A+B+C+D+E+F+G+H}$$

From previous question

$$\text{Pseudo nMos No12} \Rightarrow g = \frac{9}{6}$$

input inverter : $g = 1$

$$G = 1 \cdot \frac{9}{6} = \frac{9}{6}$$

$$H = 6 \cdot \frac{9}{6} = 9$$

$$F = GH = \frac{9}{6} \cdot 6 = 9$$

Since there are 2 stages

$$F = \sqrt{F} = \sqrt{9}$$

$$P = \frac{4n+2}{9} = \frac{4(8)+2}{9} = \frac{34}{9}$$

$$P = 1 + \frac{34}{9} = \frac{43}{9}$$

$$D = 12\sqrt{9} + \frac{43}{9} = \boxed{9.29}$$

1) unbooted dynamic 3-input NAND

Question 5) $\frac{3 \cdot 12}{3} = 12$

precharge pmos : $w_p = 1$
each nmos : $w_n = 3$

$C_{in} = 3$
 $g_d = \frac{3}{3} = 1$
 $g_{\text{out}} =$

2) Footed dynamic 3-input NAND

$4 \cdot \frac{12}{w_n} = 12 \Rightarrow w_n = 4$

Precharge pmos : $w_p = 1$
each input nmos $w_n = 4$
footer nmos $w_p = 4$

$g_d = \frac{4}{3}$

3) un footed dynamic 3-input NOR

$w_p = 1$
 $w_n = 1$

$C_{in} = 1$
 $g_d = \frac{1}{3}$

4) footed dynamic 3-input NOR

$2 \cdot \frac{12}{w_n} = 12 \Rightarrow w_n = 2$

Precharged pmos : $w_p = 1$
each input nmos $w_n = 2$
footer nmos $w_p = 2$

$g_d = \frac{2}{3}$

Question 6)

$$a) T_c \geq t_{pcq} + t_{pd,logic} + t_{setup}$$

with $T_c = 500 \text{ ps}$

$$t_{pd,logic}^{\max} = 500 - t_{pcq} - t_{setup}$$

$$b) \frac{T_c}{2} \geq t_{pdq} + t_{pd,logic} + t_{setup}$$

$$t_{pd,logic}^{\max} = 250 - t_{pdq} - t_{setup}$$

$$c) T_c \geq \max(t_{pdq} + t_{pd}, t_{pcq} + t_{pd} + t_{setup} - t_{pw})$$

$$t_{pd}^{\max} = 500 - \max(t_{pdq}, t_{pcq} + t_{setup} - t_{pw})$$

Question 7)

$$a) t_{cd,logic}^{\min} = t_{hold} - t_{ccq}$$

$$b) t_{cd,logic}^{\min} = t_{hold} - t_{ccq}$$

$$c) t_{cd,logic}^{\min} = t_{hold} - t_{ccq} + 80$$

$$d) t_{cd,logic}^{\min} = t_{hold} - t_{ccq} + 80 \text{ ps}$$

Question 8)

a) $t_{\text{borrow,max}} = 0 \text{ ps}$

b) $\frac{T_c}{2} = 250 \text{ ps}$

$t_{\text{borrow,max}} = 250 \text{ ps}$

c) $t_{\text{borrow,max}} = T_c - t_{\text{nonoverlap}}$
 $= 250 - 60 = 190 \text{ ps}$
 $\Phi = 190 \text{ ps}$

d) $t_{\text{borrow,max}} = 80 \text{ ps}$

Question 9)

a) 50 ps

b) $250 + 50 = 300 \text{ ps}$

c) $250 - 60 = 190 \text{ ps}$
 $19 + 50 = 240 \text{ ps}$

d) $80 + 50 = 130 \text{ ps}$

Question 10)

$$a) \Delta_1 = 550 \quad \Delta_2 = 580 \quad \Delta_3 = 450 \quad \Delta_4 = 200$$

$$1) T = 1200 \text{ ps}$$

$$\text{Half cycle} = 600 \text{ ps}$$

$$L1: \quad q_1 = 550$$

$$\text{window} : [600, 1200]$$

since $550 < 600 \Rightarrow L1$ does not borrow
output from L1 can continue at

$$t_1 = \max(550, 600) = 600$$

L2

$$q_2 = 600 + 580 = 1180$$

$$\text{Window} : [1200, 1800]$$

since $1180 < 1200$, L2 does not borrow

$$t_2 = \max(1180, 1200) = 1200$$

L3

$$q_3 = 1200 + 450 = 1650$$

$$\text{Window} : [1800, 2400]$$

since $1650 < 1800$, L3 does not borrow

$$t_3 = \max(1650, 1800) = 1800$$

~~L4~~ L4

$$q_4 = 1800 + 200 = 2000$$

$$\text{window} [2400, 3000]$$

since $2000 < 2400$ L4 does not borrow

~~Borrowing does not~~
~~Setup violations. None~~

$$2) T = 1000 \text{ ps}$$

Half cycle = 500 ps

L1

$$q_1 = 550$$

window [500, 1000]

since $550 > 500$ L1 borrows

$$550 - 500 = 50 \text{ ps}$$

$$t_1 = 550 \text{ ps}$$

L2

$$q_2 = 550 + 580 = 1130$$

window: [1000, 1500]

since $1130 > 1000$ L2 borrows

$$1130 - 1000 = 130 \text{ ps}$$

$$t_2 = 1130$$

L3

$$q_3 = 1130 + 450 = 1580$$

window: [1500, 2000]

since $1580 > 1500$ L3 borrows

$$1580 - 1500 = 80 \text{ ps}$$

$$t_3 = 1580$$

L4

$$q_4 = 1580 + 200 = 1780$$

window: [2000, 2500]

since $2000 > 1780$ L4 does not

borrow

$$3) T = 800 \text{ ps}$$

Half cycle = 400 ps

L1

$$q_1 = 550$$

window: [400, 800]

~~since~~ since $550 > 400$ L1 borrows

$$550 - 400 = 150 \text{ ps}$$

$$t_1 = 550$$

L2

$$q_2 = 550 + 580 = 1130$$

window [800, 1200]

since $1130 > 800$ L2 borrows

$$1130 - 800 = 330 \text{ ps}$$

$$t_2 = 1130$$

L3

$$q_3 = 1130 + 450 = 1580$$

window: [1200, 1600]

since $1580 > 1200$ L3 borrows

~~since~~ ~~1580~~ ~~> 1200~~

$$1580 - 1200 = 380 \text{ ps}$$

$$t_3 = 1580$$

L4

$$q_4 = 1580 + 200 = 1780$$

window: [1600, 2000]

since $1780 > 1600$

L4 borrows

$$1780 - 1600 = 180 \text{ ps}$$

b) $\Delta_1 = 300$, $\Delta_2 = 600$, $\Delta_3 = 400$, $\Delta_4 = 550$

1) $T = 1200 \text{ ps}$

Half cycle = 600 ps

L1

$$a_1 = 300$$

window: $[600, 1200]$

no borrowing

$$t_1 = 600$$

L2

$$a_2 = 600 + 600 = 1200$$

window: $[1200, 1800]$

no borrowing

$$t_2 = 1200$$

L3

$$a_3 = 1200 + 400 = 1600$$

window: $[1800, 2400]$

no borrowing

$$t_3 = 1800$$

L4

$$a_4 = 1800 + 550 = 2350$$

window: $[2400, 3000]$

no borrowing

2) $T = 1000 \text{ ps}$

Half cycle = 500 ps

$$a_1 = 300$$

window: $[500, 1000]$

no borrowing

$$t_1 = 500$$

L2

$$a_2 = 500 + 600 = 1100$$

window: $[1000, 1500]$

borrow

$$1100 - 1000 = 100 \text{ ps}$$

$$t_2 = 1100$$

L3

$$a_3 = 1100 + 400 = 1500$$

window: $[1500, 2000]$

$$t_3 = 1500$$

L4

$$a_4 = 1500 + 550 = 2050$$

window: $[2000, 2500]$

borrow:

$$2050 - 2000 = 50 \text{ ps}$$

$$3) T = 800ps$$

$$\text{Half cycle} = 400ps$$

L1

$$q_1 = 300$$

$$\text{window: } [400, 800]$$

no borrowing

$$t_1 = 400$$

L2

$$q_2 = 400 + 600 = 1000$$

$$\text{Window: } [800, 1200]$$

borrow

$$1400 - 1200 = 200ps$$

$$t_2 = 1000$$

L3

$$q_3 = 1000 + 400 = 1400$$

$$\text{Window: } [1200, 1600]$$

borrow

$$1400 - 1200 = 200ps$$

$$t_3 = 1400$$

L4

$$q_4 = 1400 + 550 = 1950$$

$$\text{Window: } [1600, 2000]$$

borrow

$$1950 - 1600 = 350ps$$

u

511

Question 11)

$$a) \Delta_1 = 300 \quad \Delta_2 = 400 \quad \Delta_3 = 200 \quad \Delta_4 = 350$$

$$2\Delta_1 = 600 \quad 2\Delta_2 = 800 \quad 2\Delta_3 = 400 \quad 2\Delta_4 = 700$$

$$\Delta_1 + \Delta_2 = 700 \quad \Delta_2 + \Delta_3 = 600 \quad \Delta_3 + \Delta_4 = 550$$

$$\frac{2}{3} (\Delta_2 + \Delta_3 + \Delta_4) = \frac{2}{3} (950) = 633.3$$

$$\frac{1}{2} (\Delta_1 + \Delta_2 + \Delta_3 + \Delta_4) = \frac{1250}{2} = 625$$

Worst case $T_{min} = 800 \text{ ps}$

Dominant constraint is

$$\Delta_2 \leq \frac{T}{2} \Rightarrow 2\Delta_2 = 800 \text{ ps}$$

$$b) \Delta_1 = 300 \quad \Delta_2 = 400 \quad \Delta_3 = 400 \quad \Delta_4 = 550$$

$$2\Delta_1 = 600 \quad 2\Delta_2 = 800 \quad 2\Delta_3 = 800 \quad 2\Delta_4 = 1100$$

$$\Delta_1 + \Delta_2 = 700 \quad \Delta_2 + \Delta_3 = 800 \quad \Delta_3 + \Delta_4 = 950$$

$$\frac{2}{3} (\Delta_2 + \Delta_3 + \Delta_4) = 900$$

$$\frac{1}{2} (\Delta_1 + \Delta_2 + \Delta_3 + \Delta_4) = 825$$

Worst case $T_{min} = 1100 \text{ ps}$

Dominant constraint is

$$\Delta_4 \leq \frac{T}{2} \Rightarrow T \geq 1100 \text{ ps}$$

$$c) D_1 = 300 \quad D_2 = 900 \quad D_3 = 200 \quad D_4 = 350$$

$$2D_1 = 600 \quad 2D_2 = 1800 \quad 2D_3 = 400 \quad 2D_4 = 700$$

$$D_1 + D_2 = 1200 \quad D_2 + D_3 = 1100 \quad D_3 + D_4 = 550$$

$$\frac{2}{3} (D_1 + D_2 + D_3) = \frac{2}{3} (1400) = 933.3$$

$$\frac{2}{3} (D_2 + D_3 + D_4) = \frac{2}{3} (1450) = 966.7$$

$$\frac{1}{2} (D_1 + D_2 + D_3 + D_4) = \frac{1750}{2} = 875$$

$$\text{Worst Case } T_{\min} = 1800 \text{ ps}$$

Dominant constraint is

$$D_2 \leq \frac{T}{2} \Rightarrow T \geq 2D_2 = 1800 \text{ ps}$$

Question 12)

a) $D_1 = 300$ $D_2 = 400$ $D_3 = 200$ $D_4 = 350$

single stage:

$$300 + 125 = 425$$

$$400 + 125 = 525$$

$$200 + 125 = 325$$

$$350 + 125 = 475$$

two stage bounds:

$$\frac{2}{3} (300 + 400 + 125) = 550$$

$$\frac{2}{3} (400 + 200 + 125) = 483.3$$

$$\frac{2}{3} (200 + 350 + 125) = 450$$

three stage bounds

$$\frac{1}{2} (300 + 400 + 200 + 125) = 512.5$$

$$\frac{1}{2} (400 + 200 + 350 + 125) = 537.5$$

Four stage bounds

$$\frac{2}{5} (300 + 400 + 200 + 350 + 125) = 550$$

so $T_{min} = 550$ ps

b) $D_1 = 300$ $D_2 = 400$ $D_3 = 400$ $D_4 = 550$

Single stage bounds

$$425, 525, 525, 675$$

two stage bounds

$$\frac{2}{3} (300 + 400 + 125) = 550$$

$$\frac{2}{3} (400 + 400 + 125) = 616.7$$

$$\frac{2}{3} (400 + 550 + 125) = 716.7$$

three stage bounds:

$$\frac{1}{2} (300 + 400 + 400 + 125) = 612.5$$

$$\frac{1}{2} \frac{2}{3} (400 + 400 + 550 + 125) = 737.5$$

Four stage bound

$$\frac{2}{5} (300 + 400 + 400 + 550 + 125) = 710$$

$$T_{\min} = 737.5 \text{ ps}$$

c) $\Delta_1 = 300$ $\Delta_2 = 900$ $\Delta_3 = 200$ $\Delta_4 = 350$

single stage bounds

$$425, 1025, 325, 475$$

two stage bounds:

$$\frac{2}{3} (300 + 900 + 125) = 883.3$$

$$\frac{2}{3} \frac{3}{2} (900 + 200 + 125) = 816.7$$

$$\frac{2}{3} \frac{3}{3} (200 + 350 + 125) = ~~762.5~~ 450$$

Three stage bound

$$\frac{2}{5} (300 + 900 + 200 + 350 + 125) = 750$$

$$T_{\min} = 1025 \text{ ps}$$